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Segmented capacitance calibration circuit applied in pure capacitor array structure

Abstract

A segmented capacitance calibration circuit applied in a pure capacitor array structure includes a first calibration unit, a second calibration unit and a selection switch which are all connected with a scaling capacitor. The first and second calibration units include at least one capacitor, and the selection switch is configured to select the first or second calibration unit to be connected to the pure capacitor array structure. When the first calibration unit is connected to the pure capacitor array structure and in parallel with the scaling capacitor, a negative error calibration is performed on the scaling capacitor. When the second calibration unit is connected to the pure capacitor array structure and in series with the scaling capacitor, a positive error calibration is performed on the scaling capacitor. The nonlinear problem caused by the precision error of the scaling capacitor in the pure capacitor array structure is solved.

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Background/Summary

FIELD OF THE INVENTION

(1) The invention relates to the technical field of integrated circuits, and particularly to a segmented capacitance calibration circuit applied in a pure capacitor array structure.

BACKGROUND OF THE INVENTION

(2) An analog-to-digital converter (ADC) plays an important role in analog circuits, and is most widely used in the successive approximation register (SAR) ADC circuit for example. The structure of SAR ADC includes resistor-resistor (pure resistor) type, capacitor-resistor (hybrid) type, and capacitor-capacitor (pure capacitor) type. Specifically, the resistor-resistor type is used less because of the high power consumption, and the capacitor-capacitor type is not often used in high-precision ADCs because of the overlarge the total capacitors area. While the capacitor-resistor type is most commonly used.

(3) Although the area for total capacitors in the capacitor-capacitor structure cannot be ignored, it is still used in some important cases because of its fast response speed and small power consumption.

During the application, a scaling capacitor is commonly added in the circuit for reducing the total capacitance.

(4) FIG. 1 shows the comparison between a pure capacitor array structure (the upper part of FIG. 1) with a scaling capacitor and an original array (the lower part of FIG. 1) without a scaling capacitor, where C_s is denoted as the scaling capacitor. Due to the existence of the scaling capacitor C_s , it is possible to make all the capacitors in the MSB (least significant bit) capacitor array not follow a $2\times$ relationship. In addition, due to the scaling capacitor C_s , all the capacitors in the LSB (least significant bit) capacitor array have the same weight as the original array. The total capacitance value of the original array is $32C$ (where C is a capacitance value preset based on the comprehensive consideration of layout design and cost, and reflects the relationship between capacitance values of capacitors). With the scaling capacitor, the total capacitance of the entire array is $31C/8+C_s$. Since the series combination of the scaling capacitor C_s and the LSB capacitor array must be connected to the left side of the MSB capacitor array, and the total capacitance of the LSB capacitor array is $C/8$, therefore,

$$(5) \frac{1}{C_s} = \frac{8}{C} - \frac{1}{2C} = \frac{15}{2C}.$$

(6) According to the above calculation, the capacitance value of the scaling capacitor C_s is $2C/15$. The total capacitance value of the entire capacitor array is $31C/8+2C/15$. Obviously, the total capacitance after adding the scaling capacitor C_s is reduced by about 8 times compared with the total capacitance of the original array. This is a very large improvement for high-precision SAR ADCs, which means that the capacitor area of the entire array is greatly improved.

(7) However, the disadvantage of adding the scaling capacitor C_s is that the capacitance value of the scaling capacitor C_s is often a non-integer multiple of C . In consideration of capacitance mismatch, the minimum capacitor in the least significant bit in the LSB capacitor array is often a non-integer multiple of C as well. Therefore, it's difficult to match the capacitance value of scaling capacitor C_s in practical application, which results in errors, and in turn, results in errors on each individual capacitor in the LSB capacitor array.

(8) Therefore, it is necessary to provide a segmented capacitance calibration circuit that can calibrate the scaling capacitor and be applicable to the pure capacitor array structure to overcome the above defects.

SUMMARY OF THE INVENTION

(9) The present invention aims to provide a segmented capacitance calibration circuit applied in a pure capacitor array structure, which can calibrate the error of the scaling capacitor in the pure capacitor array with high precision, and solve the nonlinear problem caused by the precision error of the scaling capacitor in the pure capacitor array structure.

(10) To achieve the above objective, the present invention provides a segmented capacitance calibration circuit applied in a pure capacitor array structure, and arranged between a most significant bit capacitor array and a least significant bit capacitor array of a pure capacitor array structure. The segmented capacitance calibration circuit includes a first calibration unit, a second calibration unit and a selection switch which are all connected with a scaling capacitor of the pure capacitor array structure, both the first calibration unit and the second calibration unit includes at least one capacitor, and the selection switch is configured to select the first calibration unit or the second calibration unit to be connected to the pure capacitor array structure. When the scaling capacitor generates a negative error, the selection switch is connected, and the first calibration unit is connected to the pure capacitor array structure and connected in parallel with the scaling capacitor so as to perform a negative error calibration on the scaling capacitor; when the scaling capacitor generates a positive error, the selection switch is disconnected, and the second calibration unit is connected to the pure capacitor array structure and connected in series with the scaling capacitor so as to perform a positive error calibration on the scaling capacitor.

(11) As a preferable embodiment, the first calibration unit includes n sets of first calibration subunits, each set of the first calibration subunits includes a first switch, and two first capacitors

that are identical, and the two first capacitors and the first switch are connected in series; when the first switch in any set of the first calibration subunits is connected, the first calibration subunit in a current set is connected in parallel with the scaling capacitor, where n denotes a number of sets for the first calibration subunits, and also denotes a number of levels for the negative error calibration, and n is a natural number greater than or equal to 1.

(12) As a preferable embodiment, a reference capacitance value is set to be twice a product of a total capacitance value of the least significant bit capacitor array and a preset calibration value; for a first set of the first calibration subunits, a capacitance value of the first capacitor in the first set is the reference capacitance value, and a capacitance value of the first capacitor in any one other set of the first calibration subunits is equal to a sum of a capacitance value of the first capacitor in a previous set of first calibration subunits and the reference capacitance value.

(13) As a preferable embodiment, the second calibration unit includes m sets of second calibration subunits, each set of the second calibration subunits includes a second switch and a second capacitor connected in series; when the second switch in any set of the second calibration subunits is connected, the second calibration subunit in a current set is connected in series with the scaling capacitor, where m denotes a number of sets for the second calibration subunits, and also denotes a number of levels for the positive error calibration, and m is a natural number greater than or equal to 1.

(14) As a preferable embodiment, a capacitance value of the scaling capacitor is set as C_s , and an error of the scaling capacitor C_s to be calibrated by an x th set of the second calibration subunits is denoted as b_x , a capacitance value of the second capacitor in the x th set of the second calibration subunits is denoted as C_x , where $1 \leq x \leq m$, and a formula for calculating the capacitance value C_x of the second capacitor is as follows:

$$(15) \frac{1}{C_s + b_x} + \frac{1}{C_x} = \frac{1}{C_s}.$$

(16) As a preferable embodiment, when the selection switch is disconnected, each first switch is disconnected.

(17) As a preferable embodiment, the selection switch, each first switch and each second switch are in a form of a transmission gate.

(18) As a preferable embodiment, the selection switch, each first switch and each second switch all includes a P-type MOS tube and an N-type MOS tube with same sizes, sources of the P-type MOS tube and the N-type MOS tube are jointly connected to form an input/output end of the first switch or the second switch, drains of the P-type MOS tube and the N-type MOS tube are jointly connected to form another input/output end of the first switch or the second switch, and gates of the P-type MOS tube and N-type MOS tube form a control end of the first switch or the second switch.

(19) In comparison with the prior arts, in the segmented capacitance calibration circuit of the present invention, a first calibration unit for the negative error calibration, and a second calibration unit positive error calibration are configured, which can be selectively connected to the pure capacitor array structure. When the first calibration unit is selected and connected in parallel with the scaling capacitor, to increase the total capacitance of the circuit, so as to offset the negative error of the scaling capacitor. When the second calibration unit is selected and connected in series with the scaling capacitor, to decrease the total capacitance of the circuit, so as to offset the positive error of the scaling capacitor. In such a way, the error calibration of the scaling capacitor is realized, and the nonlinear problem caused by the precision error of the scaling capacitor in the pure capacitor array is solved.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The accompanying drawings facilitate an understanding of the various embodiments of this

invention. In such drawings:

- (2) FIG. 1 is a schematic diagram of a conventional pure capacitor array structure;
- (3) FIG. 2 is a schematic diagram of a segmented capacitance calibration circuit applied in a pure capacitor array structure according to one embodiment of the present invention;
- (4) FIG. 3 is a schematic diagram of a switch structure in the segmented capacitance calibration circuit applied in a pure capacitor array structure according to one embodiment of the present invention; and
- (5) FIG. 4 is a schematic diagram of a segmented capacitance calibration circuit applied in a pure capacitor array structure according to another embodiment of the present invention.

DETAILED DESCRIPTION OF ILLUSTRATED EMBODIMENTS

(6) Referring to the accompanying drawings, the embodiments of the present invention are described. Similar reference numerals in the drawings represent similar elements. As described above, the present invention provides a segmented capacitance calibration circuit applied in a pure capacitor array structure, which can calibrate the error of the scaling capacitor in the pure capacitor array with high precision, and solve the nonlinear problem caused by the precision error of the scaling capacitor in the pure capacitor array structure.

(7) FIG. 2 is a schematic diagram of a segmented capacitance calibration circuit applied in a pure capacitor array structure according to one embodiment of the present invention. As shown, the segmented capacitance calibration circuit is arranged between a most significant bit (MSB) capacitor array and a least significant bit (LSB) capacitor array of a pure capacitor array structure. Specifically, in FIG. 2, the capacitor C_{msb} is equivalent to the MSB capacitor array and the capacitor C_{lsb} is equivalent to the LSB capacitor array. The calibration circuit includes a first calibration unit, a second calibration unit and a selection switch S_c which are all connected with a scaling capacitor C_s of the pure capacitor array structure. Both the first calibration unit and the second calibration unit include at least one capacitor. The selection switch is configured to select the first calibration unit or the second calibration unit to be connected to the pure capacitor array structure. When the scaling capacitor C_s generates a negative error, the selection switch S_c is connected (in turned-on state), and the first calibration unit is connected to the pure capacitor array structure and connected in parallel with the scaling capacitor C_s so as to perform a negative error calibration on the scaling capacitor C_s . When the scaling capacitor C_s generates a positive error, the selection switch is disconnected (in turn-off state), and the second calibration unit is connected to the pure capacitor array structure and connected in series with the scaling capacitor C_s so as to perform a positive error calibration on the scaling capacitor C_s . The connection/disconnection of the selection switch S_c can be controlled by an external circuit and is not discussed in detail in the present invention.

(8) By this token, when the scaling capacitor C_s generates a negative error, the selection switch S_c is connected, so that the first calibration unit is connected to the pure capacitor array structure and connected in parallel with the scaling capacitor C_s . Furthermore, the first calibration unit includes at least one capacitor, that is, the scaling capacitor C_s is connected in parallel with at least one capacitor, so that the total capacitance value in the circuit is increased to offset the negative error of the scaling capacitor C_s . When the scaling capacitor C_s generates a positive error, the selection switch S_c is disconnected, so that the second calibration unit is connected to the pure capacitor array structure and connected in series with the scaling capacitor C_s . Furthermore, the second calibration unit includes at least one capacitor, that is, the scaling capacitor C_s is connected in series with at least one capacitor, so that the total capacitance value in the circuit is decreased to offset the positive error of the scaling capacitor C_s . In such a way, the error calibration of the scaling capacitor C_s is realized, and the nonlinear problem caused by the precision error of the scaling capacitor C_s in the pure capacitor array is solved.

(9) Specifically, referring to FIG. 2, the first calibration unit in the segmented capacitance calibration circuit includes n sets of first calibration subunits, each set of the first calibration

subunits includes a first switch S_n ($S_{n1}, S_{n2} \dots S_{nn}$) and two identical first capacitors C_{n1} ($C_{n11}, C_{n21} \dots C_{nn1}$) and C_{n2} ($C_{n12}, C_{n22} \dots C_{nn2}$). The two first capacitors C_{n1} and C_{n2} , and the first switch S_n are connected in series. When the first switch of any set of the first calibration subunits is connected, the first calibration subunit in the current set is connected in parallel with the scaling capacitor C_s , where n denotes a number of sets for the first calibration subunits, and also denotes a number of levels for the negative error calibration. In the present invention, each first calibration subunit is provided with two identical first capacitors C_{n1} and C_{n2} connected in series, that's because an equivalent capacitor with relatively smaller capacitance value will be obtained after the two capacitors are connected in series, while a single capacitor with a small capacitance value will lead to a large mismatch coefficient of the entire circuit to reduce the calibration accuracy. Therefore, in the present invention, an equivalent capacitor with relatively small capacitance value is obtained by connecting two first capacitors C_{n1} and C_{n2} in series, thereby solving the problem of mismatch to realize the calibration of scaling capacitor C_s . Specifically, n is a natural number greater than or equal to 1. Taking the first set of the first calibration subunits as an example, the first calibration subunit includes the first switch S_{n1} , and two first capacitors C_{n11} and C_{n12} . The two first capacitors C_{n11} and C_{n12} are connected in series with the first switch S_{n1} . When the first switch S_{n1} is connected, the first set of the first calibration subunits is connected in parallel with the scaling capacitor C_s , that is, the first capacitor C_{n11} and C_{n12} in series connection are connected in parallel with the scaling capacitor C_s , and then connected to the pure capacitor array circuit, so that the total capacitance value connected in the circuit is increased so as to offset the negative error of the scaling capacitor C_s . Specifically, a capacitance value of the minimum capacitor (LSB capacitor, the capacitor with the capacitance value of $c/8$ as shown in FIG. 1) in the LSB capacitor array is set as k , a calibration accuracy factor is set as a , and a reference capacitance value g is set as twice the product of the calibration accuracy factor a and the capacitance value k of the LSB capacitor, that is, $g=2k*a$. For the first set of the first calibration subunits, the capacitance value of the first capacitors C_{n11} and C_{n12} in the first set is the reference capacitance value, that is, the capacitance value of the first capacitors C_{n11} and C_{n12} is denoted as g . The capacitance value of the first capacitors C_{n1} and C_{n2} in any one other set of the first calibration subunits is the sum of the capacitance value of the first capacitors C_{n1} and C_{n2} in a previous set of the first calibration subunits and the reference capacitance value g . In other words, the capacitance value of the first capacitors C_{n21} and C_{n22} in the second set of the first calibration subunits is $2g$, and the capacitance value of the first capacitors C_{n31} and C_{n32} in the third set is $3g$, . . . and the capacitance value of the first capacitors C_{nn1} and C_{nn2} in the n th set is ng .

(10) Further, referring to FIG. 4, as an example with four sets of the first calibration subunits (that is, $n=4$), a negative error calibration with the first calibration unit is performed on the scaling capacitor C_s as follows. When the selection switch S_c and the first switch S_{41} are both connected, and the first switches S_{42} , S_{43} and S_{44} are disconnected, the scaling capacitor C_s is equivalently connected in parallel with a capacitor of $0.5g$ capacitance, and the total capacitance is increased by $0.5g$ (namely the capacitance after the first capacitors C_{n11} and C_{n12} are connected in series). When the selection switch S_c and the first switch S_{42} are both connected, and the first switches S_{41} , S_{43} and S_{44} are disconnected, the scaling capacitor C_s is connected in parallel with $1g$ capacitance (namely the capacitance after the first capacitors C_{n21} and C_{n22} are connected in series), and the total capacitance is increased by $1g$. When the selection switch S_c and the first switch S_{43} are both connected, and the first switches S_{41} , S_{42} and S_{44} are disconnected, the scaling capacitor C_s is equivalently connected in parallel with a capacitor of $1.5g$ capacitance (the capacitance after the first capacitors C_{n31} and C_{n32} are connected in series), and the total capacitance is increased by $1.5g$. When the selection switch S_c and the first switch S_{44} are connected, and the first switches S_{41} , S_{42} and S_{43} are disconnected, the scaling capacitor C_s is capacitance connected in parallel with a capacitor of $2g$ capacitance (the capacitance after the first capacitors C_{n41} and C_{n42} are connected in series), and the total capacitance is increased by $2g$. To

sum up, in this first calibration unit with the four sets of first calibration subunits, the calibration range of the capacitance value is from $a \cdot k$ to $4a \cdot k$, and the accuracy of the calibrated capacitance value is $a \cdot k$. Obviously, the more the first calibration subunit of the scaling capacitor C_s in parallel (namely, the larger the value of n), the larger the calibration range, with an unchanged accuracy of the calibration. Therefore, it's optional to set an appropriate value of n according to the actual demands, to obtain the appropriate calibration range, which is not limited in the present invention.

(11) In addition, as shown in FIG. 2, in the present invention, the second calibration unit includes m sets of second calibration subunits, where m is a natural number greater than or equal to 1. Each set of the second calibration subunits includes a second switch S_m ($S_{m1}, S_{m2} \dots S_{mm}$) and a second capacitor C_m ($C_{m1}, C_{m2} \dots C_{mm}$). The second switch S_m is connected in series with the second capacitor C_m . When the second switch S_m in any set of the second calibration subunits is connected, the second calibration subunit in the current set is connected in series with the scaling capacitor C_s , where m denotes a number of sets for the second calibration subunits, and also denotes a number of levels for the positive error calibration, and m is a natural number greater than or equal to 1. Taking the first set of the second calibration subunits as an example, the second calibration subunit includes a second switch S_{m1} and a second capacitor C_{m1} . When the second switch S_{m1} is connected, the first set of the second calibration subunits is connected in parallel with the scaling capacitor C_s . That is, the second capacitor C_{m1} and the scaling capacitor C_s are connected in series and then connected to the pure capacitor array circuit, so that the total capacitance value connected in the circuit is reduced so as to offset the positive error of the scaling capacitor C_s . Specifically, the capacitance value of the scaling capacitor C_s is denoted as C_s , the error of the scaling capacitor C_s to be calibrated by the x th set of the second calibration subunits is denoted as b_x , the capacitance value of the second capacitor C_{mx} of the x th set of the second calibration subunits is denoted as C_x (where $1 \leq x \leq m$), and the formula for calculating the capacitance value C_x of the second capacitor C_{mx} of any set of the second calibration subunits is as follows:

$$(12) \quad \frac{1}{C_s + b_x} + \frac{1}{C_x} = \frac{1}{C_s} \quad (1)$$

(13) By this token, the error calibration of the scaling capacitor C_s can be carried out based on a preset error, when the second capacitor C_m is selected with an appropriate capacitance value in each set of second calibration subunits.

(14) Specifically, referring to FIG. 4, as an example, the error calibration process with the second calibration unit is explained. Since the capacitance value k of the scaling capacitor C_s is approximate to that of the LSB capacitor in the LSB capacitor array, thus it's presumed that $C_s = k$ in the calibration process, and the reference calibration error is set as h . It's understood that, the error in the first level of the calibration is the reference calibration error, namely the reference calibration error $h = b_x / x = b_1$. As an example, four sets (namely, $m = 4$) of second calibration subunits are set, and the positive error calibration on the scaling capacitor C_s by the second calibration unit is explained as follows. Presumed that, the error of each level is $h, 2h, 3h, 4h$ respectively, the sum of the capacitance value of the corresponding scaling capacitor C_s and the error capacitance value is $k + h, k + 2h, k + 3h$, and $k + 4h$ respectively. Specifically, when the error of the first level is h (i.e. $h = b_1, x = 1$), the second switch S_{m1} is connected, and the selection switch S_c and the second switches S_{m2}, S_{m3} and S_{m4} are disconnected, thus the capacitor with the capacitance value of $k + h$ is connected in series with the second capacitor C_{m1} with the capacitance value of k/h (calculated by formula (1)) to form the capacitor in the circuit, and the capacitance value of the equivalent capacitor obtained after the series connection is $1/[1/(k + h) + h/k]$. When the error of the second level is $2h$ (i.e. $b_2, x = 2$), the second switch S_{m2} is connected, and the selection switch S_c and the second switches S_{m1}, S_{m3} and S_{m4} are disconnected, thus the capacitor with the capacitance value of $k + 2h$ is connected in series with the second capacitor C_{m2} with the capacitance value of $k/2h$ (calculated by formula (1)) to form the capacitor in the circuit, and the capacitance value of the

equivalent capacitor obtained after the series connection is $1/[1/(k+2h)+2h/k]$. When the error of the third level is $3h$ (i.e. b_3 , $x=3$), the second switch Sm_3 is connected, and the selection switch Sc and the second switches Sm_1 , Sm_2 and Sm_4 are disconnected, thus the capacitor with the capacitance value of $k+3h$ is connected in series with the second capacitor Cm_3 with the capacitance value of $k/3h$ (calculated by formula (1)) to form the capacitor in the circuit, and the capacitance value of the equivalent capacitor obtained after the series connection is $1/[1/(k+3h)+3b/k]$. When the error of the fourth level is $4h$ (i.e. b_4 , $x=4$), the second switch Sm_4 is connected, and the selection switch Sc and the second switches Sm_1 , Sm_2 and Sm_3 are disconnected, thus the capacitor with the capacitance value of $k+4h$ is connected in series with the second capacitor Cm_4 with the capacitance value of $k/4h$ (calculated by formula (1)) to form the capacitor in the circuit, and the capacitance value of the equivalent capacitor obtained after the series connection is $1/[1/(k+4h)+4b/k]$. After the above calculations of the four sets of second calibration subunits, it can be seen that the capacitance value of the equivalent capacitor obtained after series connection in any set is less than k , and the value of k is depended on the specific value of x (namely the selected level for the calibration) which is corresponding with the error value size of the scaling capacitor C_s . Therefore, by connecting the second calibration subunit corresponding to the current calibration level to the pure capacitor array structure, the capacitance error of the scaling capacitor C_s can be calibrated within the range of $H-m*h$, and the capacitance accuracy of the calibration is h . Obviously, the more the second calibration subunit of the scaling capacitor C_s in series (the larger the value of m), the larger the calibration range, with an unchanged accuracy of the calibration. Therefore, it's optional to set an appropriate value of m according to the actual demands, to obtain the appropriate calibration range, which is not limited in the present invention.

(15) In addition, since each of the first calibration subunits is in parallel connection with the scaling capacitor C_s , and each of the second calibration subunits is in series connection with the scaling capacitor C_s , when the selection switch is disconnected so that the second calibration unit is used, all of the first switches Sn (Sn_1 , $Sn_2 \dots Sn_n$) are disconnected, thereby avoiding influence on the second calibration subunit.

(16) As a preferred embodiment of the present invention, all switches, including the selection switch, each first switch and each second switch, are switches in the form of transmission gates to solve the nonlinear problem caused by charge injection. As shown in FIG. 3, each switch includes a P-type MOS tube and an N-type MOS tube with the same size. The sources of the P-type MOS tube and the N-type MOS tube are jointly connected to form an input/output end V_{in} of the switch, the drains of the P-type MOS tube and the N-type MOS tube are jointly connected to form another input/output end V_{out} of the switch, and the gates (CT, CTN) of the P-type MOS tube and N-type MOS tube form the control end of the switch. In addition, in the present invention, each switch may be manually controlled by the operators one by one, or by a peripheral control circuit. Specific control modes are not limited in the present invention, as long as the switches at each level can be controlled one by one.

(17) The above-mentioned embodiments only represent several embodiments of the present invention, and the descriptions thereof are relatively specific and detailed, but should not be construed as limiting the scope of the patent invention. It should be pointed out that for those skilled in the art, several modifications and improvements can be made without departing from the concept of the present invention, which all belong to the protection scope of the present invention. Therefore, the scope of protection of the patent of the present invention shall be subject to the appended claims.

Claims

1. A segmented capacitance calibration circuit applied in a pure capacitor array structure, arranged between a most significant bit capacitor array and a least significant bit capacitor array of a pure

capacitor array structure, wherein the segmented capacitance calibration circuit comprises a first calibration unit, a second calibration unit and a selection switch which are all connected with a scaling capacitor of the pure capacitor array structure, both the first calibration unit and the second calibration unit comprises at least one capacitor, and the selection switch is configured to select the first calibration unit or the second calibration unit to be connected to the pure capacitor array structure; when the scaling capacitor generates a negative error, the selection switch is connected, and the first calibration unit is connected to the pure capacitor array structure and connected in parallel with the scaling capacitor so as to perform a negative error calibration on the scaling capacitor; when the scaling capacitor generates a positive error, the selection switch is disconnected, and the second calibration unit is connected to the pure capacitor array structure and connected in series with the scaling capacitor so as to perform a positive error calibration on the scaling capacitor.

2. The segmented capacitance calibration circuit applied in a pure capacitor array structure according to claim 1, wherein the first calibration unit comprises n sets of first calibration subunits, each set of the first calibration subunits comprises a first switch, and two first capacitors that are identical, and the two first capacitors and the first switch are connected in series; when the first switch in any set of the first calibration subunits is connected, the first calibration subunit in a current set is connected in parallel with the scaling capacitor, where n denotes a number of sets for the first calibration subunits, and also denotes a number of levels for the negative error calibration, and n is a natural number greater than or equal to 1.

3. The segmented capacitance calibration circuit applied in a pure capacitor array structure according to claim 2, wherein a reference capacitance value is set to be twice a product of a total capacitance value of the least significant bit capacitor array and a preset calibration value; for a first set of the first calibration subunits, a capacitance value of the first capacitor in the first set is the reference capacitance value, and a capacitance value of the first capacitor in any one other set of the first calibration subunits is equal to a sum of a capacitance value of the first capacitor in a previous set of first calibration subunits and the reference capacitance value.

4. The segmented capacitance calibration circuit applied in a pure capacitor array structure according to claim 2, wherein the second calibration unit comprises m sets of second calibration subunits, each set of the second calibration subunits comprises a second switch and a second capacitor connected in series; when the second switch in any set of the second calibration subunits is connected, the second calibration subunit in a current set is connected in series with the scaling capacitor, where m denotes a number of sets for the second calibration subunits, and also denotes a number of levels for the positive error calibration, and m is a natural number greater than or equal to 1.

5. The segmented capacitance calibration circuit applied in a pure capacitor array structure according to claim 4, wherein a capacitance value of the scaling capacitor is set as C_s , and an error of the scaling capacitor C_s to be calibrated by an xth set of the second calibration subunits is denoted as b_x , a capacitance value of the second capacitor in the xth set of the second calibration subunits is denoted as C_x , where $1 \leq x \leq m$, and a formula for calculating the capacitance value C_x of the second capacitor is as follows: $\frac{1}{C_s + b_x} + \frac{1}{C_x} = \frac{1}{C_s}$.

6. The segmented capacitance calibration circuit applied in a pure capacitor array structure according to claim 4, wherein when the selection switch is disconnected, each first switch is disconnected.

7. The segmented capacitance calibration circuit applied in a pure capacitor array structure according to claim 4, wherein the selection switch, each first switch and each second switch are in a form of a transmission gate.

8. The segmented capacitance calibration circuit applied in a pure capacitor array structure according to claim 7, wherein the selection switch, each first switch and each second switch are all comprises a P-type MOS tube and an N-type MOS tube with same sizes, sources of the P-type

MOS tube and the N-type MOS tube are jointly connected to form an input/output end of the first switch or the second switch, drains of the P-type MOS tube and the N-type MOS tube are jointly connected to form another input/output end of the first switch or the second switch, and gates of the P-type MOS tube and N-type MOS tube form a control end of the first switch or the second switch.
